From Images to Robot Coordinates: Dataset Design, Detector Selection and Monocular 3D Localisation for Automated EV Battery Disassembly

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Abstract—Recycling electric vehicle (EV) batteries is still largely a manual job, which is slow, expensive and genuinely dangerous. Automating it starts with perception: a robot cannot unbolt or lift a battery module until it knows where that module is, in metres, in its own coordinate frame. This paper works through that problem end to end, and reports what actually mattered. First, we show that the accuracy figures usually quoted for battery detectors are optimistic: a detector scoring 0.818 mAP@50 on the single facility it was trained on drops to 0.277 on a 225-image benchmark spanning ten independent pack sources, a 66% relative loss. Second, we assemble a multi-source training set (4,425 images, 13 sources) and compare a conventional from-scratch detector against one built on a frozen self-supervised backbone. The transformer-based model reaches 0.502 mAP@50 against 0.410, but a per-class breakdown shows the two are indistinguishable on consistently annotated packs; the entire advantage is robustness when the annotation convention shifts. Third, we find that annotation convention, not image difficulty, dominates performance: per-source module accuracy spans 0.995 to 0.043, and the worst case is a labelling disagreement rather than a hard image. Finally, we close the gap from pixels to metres without a depth camera. A fiducial-referenced monocular transform recovers metric coordinates from a single RGB view; evaluated in simulation on a URDF model of a real pack over 30 randomised trials, it achieves 5.67 mm mean 3D error at a 96.7% success rate, with joint multi-marker estimation cutting error by 75.6% over a single marker. We release the annotation set (16,945 polygon masks) and all evaluation code.

Index Terms—robotic disassembly, electric vehicle batteries, circular economy, object detection, domain generalisation, monocular 3D localisation, fiducial markers, dataset quality

I. Introduction

Electric vehicles are reaching end of life in large numbers, and their battery packs are both a hazard and an asset. A pack contains lithium, cobalt and nickel, all classified by the European Commission as critical raw materials with significant supply-chain risk [1], [2]. Recovering them matters economically and environmentally: recycling capacity demand in Europe is projected to reach 170 GWh by 2035 [3]. At the same time, EU Battery Regulation 2023/1542 requires documented condition data across the battery lifecycle [4], which manual inspection struggles to deliver consistently.

Yet disassembly today is still mostly people with hand tools. It is slow, it does not scale, and it exposes workers to high voltage and the risk of thermal runaway. This is precisely the kind of task robots should be doing, and a substantial research effort is now directed at automating it [5], [7], [8].

Every such system starts with perception. Before a manipulator can unbolt, grasp or lift anything, it has to find the battery modules and the busbars that connect them. The standard approach is a supervised object detector trained on images collected at one facility or one experimental rig.

That practice hides a problem. A recycling line does not receive one pack type; it receives whatever arrives that day, from different manufacturers, in different states of disassembly, under whatever lighting the building has. A detector validated only on the distribution it was trained on tells you very little about the next pack through the door. In our own earlier work we built exactly such a single-facility system [13], and its most important limitation was that we could not say whether it would transfer. This paper answers that question.

A second problem follows immediately, and it is the one that matters for robotics. Even a perfect detector returns pixels, and a manipulator needs metres. That gap is normally bridged with an RGB-D camera, which adds cost, adds calibration work, and brings a failure mode of its own: the specular, dark and highly reflective surfaces inside a battery pack are exactly the conditions under which consumer depth sensors degrade [20]. We show the bridge can instead be built from one RGB camera and a few planar markers placed once in the workcell.

We make four contributions:

1) We quantify how badly single-facility evaluation overstates accuracy for EV battery module detection: 0.818 → 0.277 mAP@50, a 66% relative loss (Section V-A).
2) We compare training strategies and architectures on identical data, showing a frozen self-supervised backbone improves cross-facility accuracy (0.502 vs 0.410 mAP@50) and isolating why: robustness to

annotation shift, not better detection of ordinary modules (Section V-B).
3) We show annotation convention is the dominant failure mode, with per-source accuracy spanning 0.995 to 0.043, and give the dataset audit that exposes it (Sections III-C and V-C).
4) We close the loop to robot coordinates with a monocular, fiducial-referenced transform needing no depth sensor: 5.67 mm mean 3D error over 30 randomised trials on a real pack model, in simulation (Section V-D).

Relation to our prior work. Our earlier paper [13] reported a CPU-deployable two-stage pipeline for a single facility, including condition grading. The present paper deliberately does not revisit condition assessment. It addresses the limitation that work identified but could not test – cross-facility transfer – and adds the metric localisation stage required to drive a manipulator. The dataset, the benchmark, the architectures compared and the localisation method are all new here.

II. Background and Related Work

A. Automating battery disassembly

Zang et al. [5] survey robotic EV battery disassembly and conclude that the process remains predominantly manual, with perception and task uncertainty as the principal obstacles. Wegener et al. [6] established early on that pack designs vary enough between manufacturers that fixed automation does not transfer, which is the practical reason perception must generalise. Tan et al. [7] demonstrate a collaborative robot that unfastens and collects screws by fusing vision with force/torque sensing, reporting over 90% unscrewing efficiency. RAPID [8] integrates open-vocabulary detection with RGB-D sensing on a gantry manipulator and reports 0.9757 mAP@50 for component detection.

RAPID's manipulation numbers are worth dwelling on, because they set the honest scope for any perception claim. With that detection accuracy, one-shot vision-driven grasping succeeded 57% of the time, rising to 83% with visual servoing and 97% with taught-in poses. Perception accuracy is necessary but plainly not sufficient. We therefore restrict our claims to perception and localisation, and report neither as a grasp success rate.

B. Perception for battery components

Most published work targets fasteners rather than modules. Screw detection has been addressed with Faster R-CNN plus rotation edge similarity [9], with YOLOX-family models [10], and in a recent YOLOv11 versus Faster R-CNN benchmark [11]. Module- and busbar-level perception is comparatively under-explored; the closest work applies instance segmentation with point-cloud registration to grasp busbars from a single module type [12].

A recurring theme is data scarcity. Every study above collected and hand-annotated its own images – 1,719 in [10] – and none released them. [11] resorts to laptop screws as a proxy domain and states plainly that the field is hampered by a lack of public datasets. That motivates our release.

C. Generalisation, and why data may matter more than architecture

Domain shift is a well-documented failure mode for detectors [14], [15]: models that appear strong on their own test split degrade sharply on data collected differently, even when the task is unchanged. Self-supervised backbones such as DINOv2 [17] learn features that transfer without task-specific fine-tuning, and detection transformers built on them, such as RF-DETR [18], are reported to generalise better than from-scratch detectors in low-data regimes. Whether that holds on the cluttered, specular imagery of battery internals is an open question we test directly.

Running alongside this is the data-centric argument [16] that label quality often limits performance more than model capacity, and that widely used benchmarks contain enough label noise to distort conclusions. Our own prior work found the same effect [13]; here we quantify it across facilities.

D. Recovering metric pose from images

Vision-guided disassembly usually obtains depth from RGB-D [8] or from point-cloud registration against a reference model [12]. Both add hardware and calibration cost, and structured-light and time-of-flight sensors are unreliable on the specular metal and dark polymer that dominate battery internals [20]. Planar fiducials such as ArUco [21] offer an alternative: a marker of known geometry fixes a metric frame from a single view via perspective-n-point [22], and is widely used for camera pose estimation and hand-eye calibration. We take this route and quantify what it costs.

III. Dataset: Construction and Analysis

A. Sources

We assembled 4,425 training images from 13 public sources spanning ten EV pack families (Table I), annotated with two classes, module and busbar. Licences differ across sources and one is non-commercial; we record per-source provenance so any single source can be excluded.

B. A cross-facility test set, and why deduplication is not optional

Reported accuracy only means something against a benchmark that resembles deployment. We therefore built a cross-facility test set of 225 images (780 instances) drawn from all sources and held out entirely from training. We verified with perceptual hashing (dHash, Hamming distance ≤ 6) that no near-duplicate crosses the train/test boundary for any source we report per-source results on.

This turned out to be necessary rather than precautionary. One candidate source, a 712-image public

TABLE I
Principal data sources (13 total, ten pack families)

Source	Imgs	Pack family
ev-battery-comp-detect.	1,045	multi-manufacturer
ev-battery-components	680	Audi/BMW/Chevy
last_exp4	483	multi-manufacturer
ev-battery-pack	435	single lab rig
battery_comp	280	Nissan Leaf
ue_d1_defect	237	module close-ups
final_mobilenet	216	multi-manufacturer
automated-disassembly	117	disassembly cell
bmw_i3, ue_rav4	207	BMW i3, RAV4
Others (3 sources)	188	mixed

collection covering 19 vehicle types, re-publishes images from another source already in our set: 78 were byte-identical duplicates of our training images. Merging without checking would have leaked training data straight into the test set and inflated every number in this paper.

C. What a dataset audit reveals

We audited all 4,425 training images for label and image quality. Of these, 23.5% carry no labels, 12.3% score low on a blur metric, 20.4% are greyscale, and none contain degenerate boxes. Two findings changed how we treated the data.

First, blur and greyscale are largely false alarms. The lowest-scoring images are smooth, closed pack lids – a Laplacian variance metric reads flat metal as "blurry" – and the greyscale images come entirely from two monochrome sources. Both add useful variation, and deleting them on the metric alone would be a mistake.

Second, the genuinely harmful category is unlabelled images that nonetheless contain visible components. Some of the 23.5% is legitimate background – a closed pack, an empty bench – and roughly 10% background is healthy because it suppresses false positives. But inspection confirmed open packs with clearly visible busbar assemblies carrying no annotations at all. Those frames actively teach the detector that busbars are background, and plausibly contribute to the weak busbar scores reported below.

We also found 16,945 of 19,522 annotations are polygon masks (5–151 vertices) rather than boxes, a richer signal than detection consumes. We convert them to enclosing boxes for evaluation, and note that an early version of our own pipeline silently discarded every polygon label, trained on 20% of the annotations, and produced an inflated score that looked entirely plausible. The failure is easy to introduce and hard to notice.

IV. Method

A. Pipeline

The system has two stages. A detector localises module and busbar instances in the image. A localisation stage then lifts each detection into metric robot-frame coordinates, so the output is directly usable by a manipulator.

B. Detectors and training strategies compared

Single-facility specialist. YOLOv8n trained only on the single-facility source, following the conventional approach and our own prior work [13].

Multi-source YOLO11n. YOLO11n (2.6 M parameters) [19] trained from COCO initialisation on all 13 sources, 150 epochs, 640 px, SGD.

Multi-source RF-DETR. RF-DETR-Nano [18], a real-time detection transformer with a frozen DINOv2 [17] backbone, fine-tuned on the same data at 384 px for 60 epochs.

C. Evaluation protocol

Confidence scores are not comparable across architectures, so every model is scored with a single metric implementation at a common low confidence threshold of 0.05, on identical images, against identical ground truth including polygon-derived boxes. We emphasise this because mixing metric implementations across architectures silently produced a spurious result in our own early experiments.

D. Monocular fiducial-referenced localisation

Let $\mathbf{K}$ be the intrinsic matrix of a calibrated monocular camera. We place N ArUco markers [21] of known side length at known positions on the workcell plane. Detecting their corners gives $4N$ correspondences between image points $\mathbf{u}_i$ and world points $\mathbf{X}_i$, from which the camera pose $(\mathbf{R}, \mathbf{t})$ follows by minimising reprojection error [22],

$$(\mathbf{R}^\star, \mathbf{t}^\star) = \arg\min_{\mathbf{R},\mathbf{t}} \sum_{i=1}^{4N} \left\| \mathbf{u}_i - \pi(\mathbf{K}, \mathbf{R}, \mathbf{t}, \mathbf{X}_i) \right\|^2, \quad (1)$$

where $\pi(\cdot)$ is the perspective projection. Using all markers jointly rather than one at a time over-constrains the fit and averages out per-marker corner noise.

Because packs rest on a support of known height, the detected module centroid $\mathbf{u}_b$ back-projects to a unique 3D point: the camera ray through $\mathbf{u}_b$ is intersected with the plane $Z = h$,

$$\mathbf{X}_b = \mathbf{o} + \lambda \mathbf{d}, \qquad \lambda = \frac{h - o_z}{d_z}, \qquad \mathbf{d} = \mathbf{R}^{\star\top} \mathbf{K}^{-1} \tilde{\mathbf{u}}_b, \quad (2)$$

with $\mathbf{o} = -\mathbf{R}^{\star\top} \mathbf{t}^\star$ the camera centre. The result is a metric grasp target in the robot frame from one RGB image. No depth buffer, segmentation mask or CAD registration is used, and no marker is ever attached to the battery: the object is found by the detector, and markers only fix the frame.

V. Results

A. How much single-facility evaluation overstates accuracy

Table II and Fig. 1 quantify the central problem. The specialist scores 0.818 mAP@50 on the single-facility benchmark it was trained and validated on, and 0.277 on the cross-facility benchmark: a 66% relative loss. Its

TABLE II
Cross-facility generalisation gap

Model	In-domain mAP@50	Cross-facility mAP@50
Single-facility specialist	0.818	0.277
Multi-source YOLO11n	0.195	0.410
Multi-source RF-DETR	0.541	0.502

TABLE III
Detector comparison, cross-facility test (225 images)

Model	mAP @50	mAP @50–95	mod.	bus.
Specialist	0.277	–	0.231	–
YOLO11n (multi-src)	0.410	0.256	0.547	0.304
RF-DETR (DINOv2)	0.502	0.312	0.680	0.353

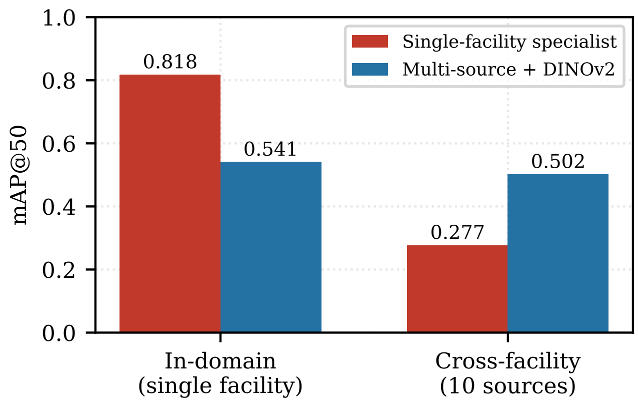

Fig. 1. A single-facility detector loses 66% of its accuracy when evaluated across ten pack sources. The multi-source model with a frozen DINOv2 backbone is far more stable across the two regimes.

in-domain number simply does not predict deployment behaviour.

The reverse also holds, and is worth stating: the multi-source models score worse in-domain than the specialist. Training for breadth costs peak accuracy on any one pack, and which trade you want depends on whether your line sees one pack type or many. The size of that drop for YOLO11n (0.195) is larger than breadth alone explains, and is not a training failure: the in-domain benchmark happens to be the one source whose annotation convention diverges from the rest, a point we return to in Section V-C. RF-DETR, evaluated on exactly the same images, retains 0.541.

B. Comparing architectures

Under the identical protocol above, RF-DETR reaches 0.502 mAP@50 against 0.410 for YOLO11n (Table III), and leads on the stricter mAP@50–95 (0.312 vs 0.256).

Taken alone that comparison is misleading. Restricting evaluation to the seven sources sharing a consistent annotation convention (177 images), the two models are effectively tied: overall 0.558 (RF-DETR) vs 0.544 (YOLO11n), module 0.771 vs 0.774, busbar 0.366 vs 0.344. The DINOv2 backbone confers no advantage on conventionally annotated packs.

The whole aggregate advantage therefore comes from behaviour under annotation shift. Adding the outlier source drops YOLO11n's module score from 0.774 to 0.547, but RF-DETR's only from 0.771 to 0.680. The claim we can defend is narrow and specific: a frozen self-supervised backbone does not detect ordinary modules better, but it degrades far more gracefully when the target definition moves. On a line receiving heterogeneous packs, graceful degradation is the property worth paying for.

C. Annotation convention as the dominant failure mode

Per-source module mAP@50 is excellent on most packs – 0.995, 0.910, 0.873, 0.749 – and collapses to 0.043 on exactly one source.

That outlier is not a harder image. It annotates small internal sub-components in extreme macro views as "modules", while every other source annotates pack-level modules. The detector learned the majority convention and applied it consistently; the collapse is a disagreement about the target, not a failure of perception.

Two consequences follow. Aggregate mAP over sources with inconsistent conventions is not meaningful, so per-source reporting is essential. And an attempt to repair the outlier automatically with an open-vocabulary grounding model failed outright: on greyscale macro imagery the proposals were unusable. Convention conflicts are not cheaply fixable after the fact, which argues for settling annotation guidelines before collection rather than after.

Busbar is the weaker class for every model (0.30–0.37), and 61% of busbar instances in the test set come from one source, with only four of eight sources annotating busbars at all – the same coverage problem in a different guise.

D. From detection to metric robot coordinates

A detection is only actionable once expressed in the robot's frame. We evaluate the localisation stage of Section IV-D in a PyBullet [23] simulation with a calibrated monocular RGB camera, in three stages of increasing realism. Ground truth is read from the simulator for validation only and never enters the estimator.

One marker versus several. Fig. 3 shows the arrangement: markers sit on the workcell surface, never on the pack itself. With a single marker (ID 0, 0.12 m) over 30 trials, detection succeeded in 100% of trials, marker localisation error was 4.05 mm mean, and the propagated error at the service point was 7.35 mm mean (median 6.97, max 15.93). Solving jointly over three markers on the same scenes cut service-point error to 1.79 mm mean (median 1.60, max 4.91) with a joint reprojection RMSE of 0.47 px – a 75.6% reduction for no extra hardware.

TABLE IV
Monocular fiducial-referenced localisation (30 trials each)

Configuration	Success rate	Mean 3D error (mm)	Median (mm)
Fixed-height baseline	–	45.65	–
Single marker	100%	7.35	6.97
Multi-marker (3)	100%	1.79	1.60
Full pack model (4)	96.7%	5.67	4.51

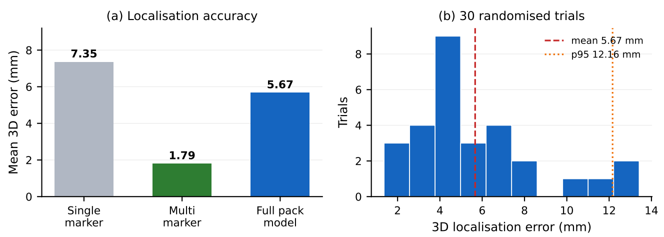

Fig. 2. Monocular localisation accuracy. (a) Joint multi-marker estimation cuts error by 75.6% over a single marker; the full pack model is harder than primitive geometry but stays under 6 mm. (b) Error distribution over 30 randomised trials.

Over-constraining the pose fit was the single most effective change we made.

A real pack model. We then replaced primitive geometry with a URDF model of a real 1S4P pack and randomised its placement over 30 trials ($x \in [0.375, 0.525]$ m, $y \in [-0.055, 0.055]$ m, yaw $\in [-25°, 25°]$). Results are in Table IV and Fig. 2. All four workspace markers were detected in 30/30 trials and the detector found the pack in 29/30, an end-to-end localisation success rate of 96.7%. Mean 3D error was 5.67 mm (median 4.51, 95th percentile 12.16), decomposing into 3.25 mm in-plane and 4.07 mm along the optical axis – the anisotropy one expects from a monocular method. Against a naive single-view baseline assuming a fixed target height (45.65 mm mean error), that is an 87.6% reduction.

The residual error budget is the most useful output here. Camera position error was 2.26 mm and workspace reprojection RMSE 0.354 px, so a substantial share of the 5.67 mm total traces to the reference frame rather than to detection. Improving marker placement and calibration is therefore a more productive target than sharpening the detector – not obvious in advance, and easy to get backwards.

Set against the manipulation literature, RAPID reports 0.9757 mAP@50 yet 57% one-shot grasp success [8], so the loss occurs between perception and actuation rather than in detection. A 5.67 mm target with a 12.16 mm 95th percentile sits inside the closing tolerance of a compliant gripper on a module of this size, suggesting the monocular route is viable for coarse positioning with final alignment handed to force feedback or visual servoing.

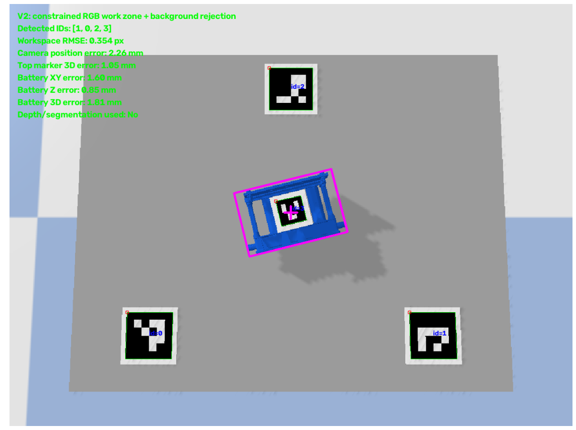

Fig. 3. Localisation scene. Workspace markers fix the metric reference frame; the pack is found by the RGB detector (magenta outline) and never carries a marker used as an object label. Workspace reprojection RMSE 0.354 px.

E. Deployment and negative results

INT8 quantisation of the detector to ONNX cut latency 20% and size 46% (44.8 ms, 3.4 MB on CPU) for a loss of 0.007 mAP@50. Localisation adds only marker detection and a PnP solve, negligible against detector inference.

Three failures constrain the design space. Open-vocabulary detection is unusable here: YOLO-World, prompted with "ev battery module" and "busbar", scored 0.004 mAP@50, battery internals being too far out of distribution despite strong general-domain results. Naive multi-source merging reduced module mAP@50 to 0.331, below the single-source baseline – more data made the model worse. And copy-paste and glare augmentation reduced mAP in our setting, despite being widely reported as beneficial elsewhere.

VI. Limitations

Our claims concern perception and localisation. We report metric accuracy, not grasp success: no physical manipulator was driven from these estimates, and as RAPID's 57% one-shot rate at 0.9757 mAP@50 shows [8], perception accuracy does not convert directly into manipulation success. Hand-eye calibration error, gripper compliance and contact dynamics are outside our scope.

The localisation results are obtained in simulation. Simulation isolates the geometry of the estimator from sensor noise, which is what we set out to measure, but it omits motion blur, rolling shutter, marker occlusion by the manipulator, and the specular highlights that make real battery internals difficult. Real-camera validation is required before these figures can be claimed for a deployed cell, and we expect some degradation. The method also assumes packs rest on a support of known height; stacked or tilted packs violate that and would need depth or a second view.

Finally, our data is assembled from public sources rather than collected under controlled conditions, so pack-type coverage is uneven and licences vary.

VII. Conclusion

Single-facility evaluation substantially overstates how well EV battery module detectors will perform in a recycling facility: a detector scoring 0.818 mAP@50 in-domain scored 0.277 across ten pack sources. Multi-source training with a frozen DINOv2 backbone raises cross-facility accuracy to 0.502 mAP@50, and we showed that this advantage is specifically robustness to annotation-convention shift rather than better detection of ordinary modules. Annotation convention proved to be the dominant and measurable failure mode, with per-source accuracy from 0.995 to 0.043 – a result that points to dataset design, not architecture, as the first thing to get right.

We then closed the loop from pixels to metres without a depth sensor. A fiducial-referenced monocular transform localised a real pack model to 5.67 mm mean 3D error at a 96.7% success rate, and solving jointly over several markers rather than one reduced error by 75.6%. Because most of the residual budget traces to the reference frame rather than to detection, calibration quality is the more productive engineering target.

Future work is real-camera validation of the localisation stage and integration with a manipulator under force feedback, moving from a metric target to a completed grasp.

Data Availability

Annotations (16,945 polygon masks), evaluation code and trained detector weights are available at https://github.com/Komil-jon/ev-battery-vision-pipeline.

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